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Implemented the First Chemical EOR Pilot in the Complex Giant Sandstone of North Kuwait: Adaptive Resilient Strategies for the challenging Harsh Environment of High Divalent Ions, Extreme Salinity, and Active Aquifer

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Abstract

The field implementation of alkaline-surfactant-polymer (ASP) flooding presents significant technical and operational challenges in heterogeneous reservoirs under harsh conditions (~280000 ppm salinity, 19000 ppm $\text{Ca}^{2+}/\text{Mg}^{2+}$ and 90°C temperature). This paper presents a field-proven and innovative ASP development strategy to address key risks, including elevated scaling, ASP formulation effectiveness, and the impact of an active aquifer in a tidally influenced, distributary channel-dominated sandstone reservoir.

An inverted five-spot pilot was implemented after the ASP formulation was designed in the laboratory and validated through a single-well chemical tracer test (SWCTT). Residual oil saturation after waterflooding was assessed using integration from special core analysis (SCAL), liquid trapper cores, time-lapse saturation logs, downhole sampling, and production/injection data. During the soft water injection phase, surveillance revealed several risks, including significant scaling at producers, failure to achieve target salinity and hardness, reduced ASP formulation effectiveness, reservoir cooling, aquifer influence, and fluid drift. To mitigate these risks, a detailed monitoring and evaluation program was implemented. This included compositional simulations, tracer injection, fluid sampling, downhole pressure and temperature measurements, pressure fall-off (PFO) testing, production and injection data analysis, and the use of various logging tools such as time-lapse saturation logs, injection logging tools (ILT), high-precision temperature (HPT), and spectral noise logging (SNL).

Field results from tracer injection and production fluid analysis revealed inconsistent reductions in salinity and hardness across producer wells, with elevated salinity and persistent divalent ion concentrations observed near wells influenced by aquifer influx. These findings prompted the optimization of the pre-flush strategy using high-resolution reservoir simulations. The revised pre-flush approach replaced the original 7 pore-volume-injection (PVI) of softened water with ~ 2 PVI of softened water followed by a ~0.8–1.0 PVI viscous polymer slug prior to ASP injection. This strategy significantly reduced scaling risks by lowering $\text{Ca}^{2+}/\text{Mg}^{2+}$ concentrations from 1,580 ppm to a manageable range of 35–158 ppm at the producer wells. It also minimized reservoir cooling and reduced the overall pilot duration by ~370 days, due to the reduced volume of softened water injected. Additionally, it enhanced ASP effectiveness, improved oil recovery, and led to a more favorable chemical-to-oil ratio. Commercially available scale scavengers were

carefully selected to mitigate unexpected scale deposition near the wellbore, ensuring sustained flow and well integrity. The selected synthetic polymer (~20% ATBS molecule AN125) demonstrated long-term thermal, chemical, and mechanical stability under harsh reservoir conditions. Furthermore, the optimized ASP formulation maintained its effectiveness across a wide temperature range (50–90°C). These results confirm the strategy's suitability for future ASP flooding deployment in challenging reservoir environments.

Keywords: Alkaline Surfactant Polymer, EOR, strategy to handle scaling, harsh environment, smart surveillance

Introduction

Even after primary and secondary recovery, a significant portion of oil remains trapped in reservoirs due to capillary forces and reservoir heterogeneity. This trapped oil includes both residual and bypassed oil, which are key targets for Enhanced Oil Recovery (EOR). Among EOR techniques, surfactant-based chemical EOR methods, such as Alkali-Surfactant-Polymer (ASP) flooding, are highly effective. ASP flooding reduces interfacial-tension (IFT) and improves mobility to mobilize both residual and bypassed oil from the reservoir. Specifically, polymers primarily enhance volumetric sweep efficiency by adjusting the mobility ratio, while surfactants improve displacement sweep efficiency by reducing IFT, thereby increasing the dimensionless capillary-number (N_c), which is defined as (El Din Saad Ibrahim, 2009; Yeganeh et al., 2016):

$$N_c = \frac{v \cdot \mu}{\sigma \cdot \cos\theta}$$

Where σ , v , θ and μ are the IFT between the displaced and displacing fluids, displaced fluid darcy velocity, contact angle and displacing fluid viscosity.

When the capillary number (N_c) is below a critical threshold, reducing residual oil saturation becomes challenging. However, once N_c exceeds this threshold, oil recovery improves significantly, particularly when N_c reaches a critical value typically associated with an interfacial tension (IFT) reduction to around 0.01 mN/m (Nguyen et al., 2005; Lv and Liu, 1984; Fulcher et al., 1985). N_c can be increased by raising the injection velocity, increasing the viscosity of the displacing fluid, or reducing the IFT. However, higher Darcy velocities may risk damaging the reservoir, and increasing fluid viscosity has practical limitations. Therefore, reducing IFT remains a viable and effective approach, commonly achieved through surfactant-based chemical EOR.

In the chemical EOR, the surfactants reduce IFT by forming a film at the oil-water interface, weakening capillary forces and allowing trapped oil to flow more easily through pore spaces. This improves oil mobility and shifts relative permeability in favor of oil flow. Effective ASP design should address both poor mobility ratios and capillary trapping. Hence, ASP flooding is considered a promising chemical EOR method, with field tests showing up to 33% additional oil recovery. It has been successfully applied in countries like China, India, the USA, and Canada, with notable performance in China's Daqing Oil Field (Gang et al., 2007; Liao et al., 2001; Jiecheng et al., 2014, 2011).

An inverted 5-spot injection pattern ASP pilot has been implemented in Kuwait's Raudhatain Zubair (RAZU) reservoir following successful laboratory evaluations and single-well chemical tracer tests. This pilot covers a 1.5-acre area and includes four producer wells, one central injector, one observation well for sampling, and an additional observation well for logging with a fiberglass casing, as illustrated in Fig. 1. The reservoir presents significant challenges for ASP deployment due to high formation temperature (~90°C), a strong underlying aquifer, very high salinity (~260,000 ppm), and elevated concentrations of divalent ions (~16,000 ppm Ca^{2+} and ~4,000 ppm Mg^{2+}). Although the reservoir has favorable rock and oil properties, these harsh conditions complicate ASP formulation effectiveness and increase the risk of inorganic scale during alkali injection.

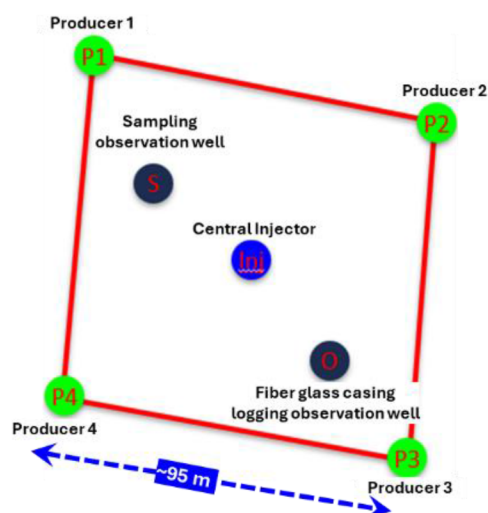


Figure 1—ASP EOR Pilot configuration and description

To address these issues, a comprehensive risk assessment was conducted using integrated surveillance data. Key observations included precipitation of elevated scale at producer wells, ineffective pre-flush performance in achieving target salinity and hardness, potential cooling impacts on ASP formulation effectiveness and aquifer-driven fluid movement toward the pilot area. Post-waterflood residual oil saturation was estimated by integrating various datasets, including SCAL results, liquid-tapered core analysis, time-lapse saturation logs, downhole fluid sampling, and production/injection data. In addition, pilot operational and performance risks were evaluated through compositional reservoir simulations, tracer tests, fluid chemistry analysis, pressure and temperature measurements, pressure fall-off (PFO) analysis, and a suite of logging tools such as time-lapse saturation logs, injection logging tools (ILT), high-precision temperature (HPT) logging, and spectral noise logging (SNL). Finally, field-proven and innovative ASP development strategies were designed to mitigate above critical risks under harsh reservoir conditions.

Real-world operational challenges in ASP flooding deployment in the field

The deployment of ASP flooding in the field, both as a pilot and for commercial development, presents several operational challenges, which are discussed below:

Scale Precipitation and Chemical Stability: ASP flooding often causes calcium carbonate scale formation due to reactions between injected alkali and divalent cations in formation water, which damage and clog equipment such as electric submersible pump (ESP) and tubing. In Oman's Marmul field, ESP failures occurred soon after ASP injection, requiring well shutdowns and chemical reformulation with scale inhibitors (Al Farsi et al., 2023). Additionally, alkali interactions with reservoir minerals can degrade chemicals and reduce flood efficiency, as seen in Malaysia's Angsi Field, necessitating mid-pilot adjustments and increasing operational complexity (Olajire, 2014).

ESP Failures and Shear Degradation: Alkali and surfactant mixtures in ASP can cause corrosion and plug in the pumps and pipelines due to polymer precipitation and scale. The Malaysia's Angsi Field faced injection pump blockages caused by poor seawater treatment and polymer hydration, which increases maintenance downtime (Othman et al., 2007). Moreover, high molecular weight polymers like HPAM are shear-sensitive, improper pump selection leads to polymer degradation, loss of viscosity and reduced efficiency. The Daqing oil fields and some oil fields in Canada showed the need for low-shear pumps and viscosity correction to maintain chemical effectiveness (Olajire, 2014).

Fines Migration and Permeability Loss: Fines migration, where tiny particles are dislodged by injected fluids, can block pore spaces and reduce permeability, especially near wellbores. This phenomenon leads

to lower production rates and often requires mitigation techniques, such as acid stimulation or nanoparticle treatments to restore flow (Shutang & Qiang, 2010).

Chemical Adsorption: Surfactants and polymers are absorbed by reservoir rocks, reducing the effective chemical concentration and thereby impairing sweep efficiency. Unexpectedly high adsorption in China's Daqing Oilfield, caused by reservoir mineralogy and temperature, led to increased chemical consumption, extended pilot durations, and higher costs, requiring adjustments to the injected chemical concentrations (Guo et al., 2017).

Reservoir Heterogeneity and Uneven Sweep: Reservoir heterogeneity such as fractures, faults, barriers/baffles and thief zones can cause uneven sweep and early injected fluid breakthrough resulting in bypass oil. In Russia's West Salym field, the pilot results underperformed, prompting enhanced reservoir characterization and conformance control measures (Volokitin et al., 2017).

Produced Fluid Separation: Surfactants in ASP formulations create stable oil-water emulsions, complicating fluid separation and increasing costs due to additional demulsifier use. In Angsi Field offshore Malaysia, persistent emulsions challenged standard separation equipment, highlighting the need for improved separation technologies and chemical treatments (Othman et al., 2007).

Space and Transport Restrictions: Offshore and remote locations pose logistical challenges for ASP operations, including limited space, lack of support infrastructure, and difficult chemical handling. The Angsi Field pilot operated without an on-site lab, had limited deck space, and relied on seawater needing softening, complicating mixing and monitoring (Othman et al., 2007). Transporting and storing large chemical volumes introduces risks of delays, degradation, spills, and challenges in maintaining chemical stability, especially in harsh environments (Olajire, 2014; Alvarado and Manrique, 2013).

Requirement for Skilled Personnel: ASP flooding requires highly trained personnel proficient in chemical EOR processes. Training delays and staff shortages can slow response times and increase safety risks. Offshore pilots particularly depend on operators skilled in chemical injection, monitoring, and safety compliance to manage complex field conditions effectively (Olajire, 2014).

Subject reservoir, EOR pilot, and simulation model descriptions

The RAZU oil field is a deep (9,000 to 9,500 ft) sandstone reservoir and a large Cretaceous-age anticlinal structure. The reservoir has good rock quality, with permeability ranging from 200 to 5,000 mD, and porosity from 16 to 30%. This reservoir has a harsh environment, including high temperature ($\sim 90^{\circ}\text{C}$), high salinity ($\sim 260,000$ ppm), elevated concentrations of divalent ions ($\sim 16,000$ ppm Ca^{2+} and $\sim 4,000$ ppm Mg^{2+}), and a strong underlying aquifer. The EOR target is the high-quality Z46CH sand, formed in tidal and estuarine channel environments. Core analysis identified two main facies in the 46CH channel: a cleaner upper sandstone (0–5% clay) and a more clay-rich, bioturbated lower unit (10–30% clay) (Khan et al., 2024).

For reservoir simulation modeling, these were grouped into clean sand, silty sand, and silty claystone using shale volume, porosity, and water saturation data, as depicted in Fig. 2. Facies distribution was modeled using the indicator kriging method, with fifty realizations run to select a base model. Porosity was simulated using sequential Gaussian simulation, while permeability was calculated from porosity and distributed conditionally using Gaussian simulation. Fig. 3 shows the permeability variation in the pilot area. To preserve vertical heterogeneity essential for fluid flow analysis, no vertical upscaling was done near the pilot area. Instead, horizontal upscaling was applied farther from the pilot using dynamic amalgamation to reduce computation time, as shown in Fig. 4. High-resolution grids near the pilot captured injection performance, while coarser grids handled surrounding flow dynamics.

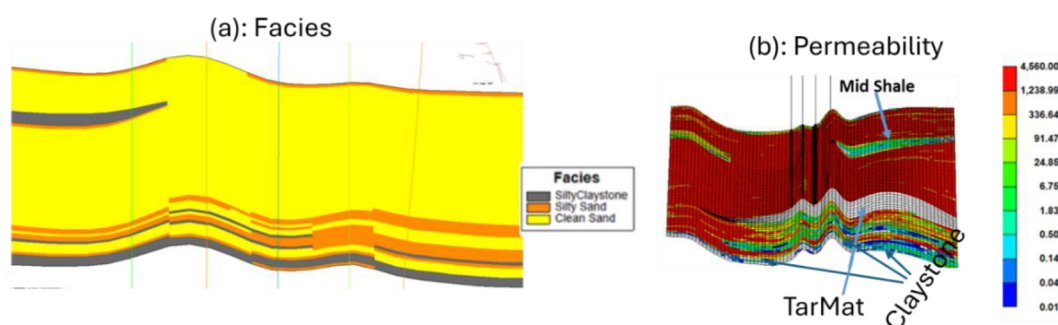


Figure 2—Reorientation of facies and TarMat in vertical cross sections

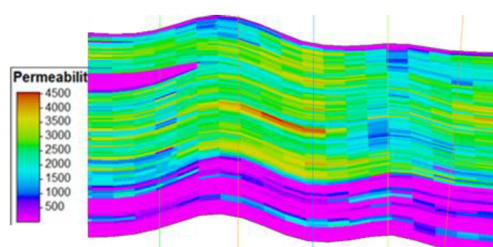


Figure 3—Permeability variation in pilot area

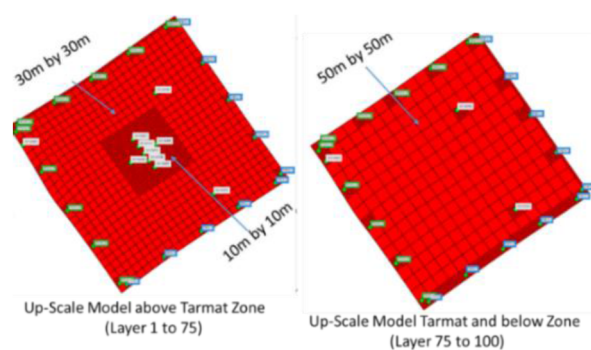


Figure 4—Simulation model upscaling strategies

Capillary pressure data from core plugs confirmed good-quality rock with low entry pressure and consistent pore sizes, as shown in Fig. 5(a). Water saturation ranged between 0.05 and 0.15 in cleaner sands, and water relative permeability endpoints were based on centrifuge tests. Since Corey exponents could not be determined experimentally in high-permeability rock, typical sandstone values (2.5–3) were used. Relative permeability curves were developed for both waterflood and ASP conditions, as shown in Fig. 5(b & c). The pilot area, currently at residual oil saturation, was initialized using recent saturation and pressure data from cores, logs, and formation tests, confirming alignment with the history-matched full-field model. The local pressure gradient is about 0.48–0.49 psi/ft, and sector boundaries were defined based on full-field flow data.

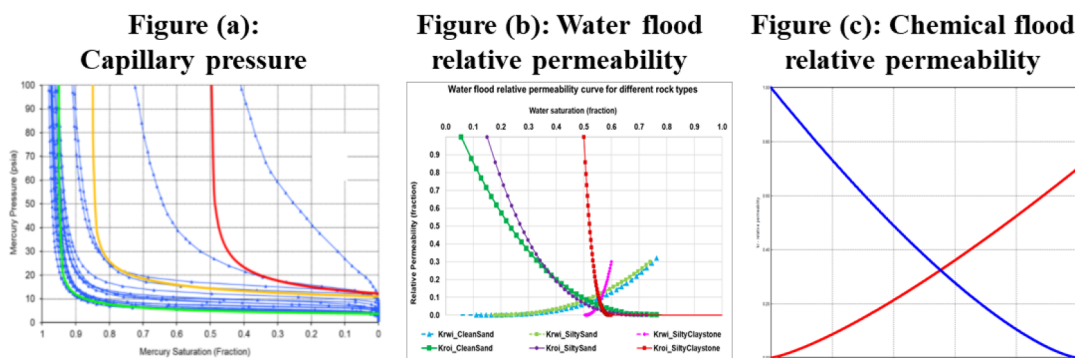


Figure 5(a)—Capillary pressure and (b) relative permeability for water flooding and (c) relative permeability for ASP flooding

Results and Discussion

Risks Analysis in ASP Pilot Field Implementation

Uncertainty in Estimation of Residual Oil Saturation After Waterflood. The open-hole logs show remaining oil saturation as 18–30% (see Fig. 6). However, a downhole sample analysed by a fluid analyser during pressure measurement at the time of drilling indicated no oil presence. Therefore, the estimated remaining oil saturation is uncertain and will be further analysed and mitigated through pilot production samples, liquid trapper core analyses, high-quality data from special-core-analysis (SCAL), time-lapse saturation logs, and single-well chemical tracer test (SWCTT).

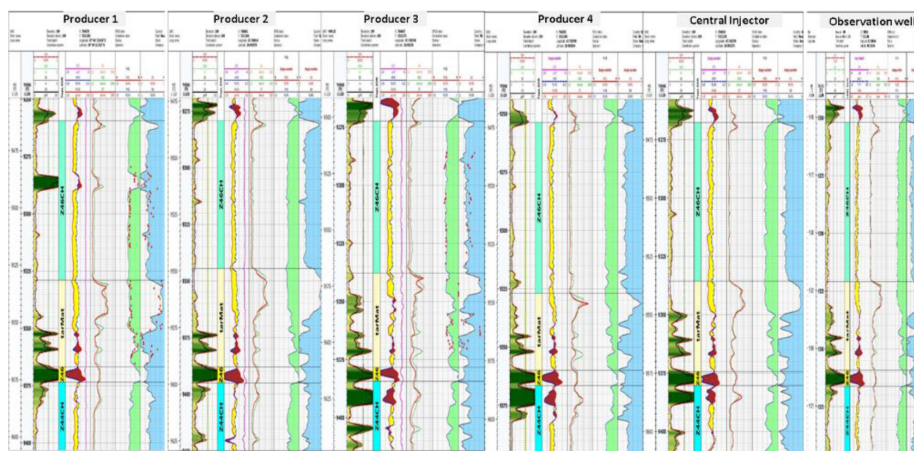
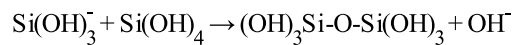
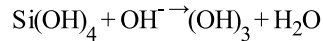


Figure 6—Open hole log saturation in newly drilled pilot wells

Scaling Potential Risk Analysis. In ASP flooding, alkali reacts with acidic compounds in crude oil to form in-situ surfactants, enhancing oil recovery. It also reduces surfactant adsorption on reservoir rocks by increasing the fluid's pH, making the rock surface more negatively charged. This improves surfactant efficiency and lowers the required concentration. For oils with high acid content (high TAN), alkali further helps by generating natural soaps through saponification, reducing IFT and enhancing oil recovery. Alkali also influences rock wettability and salinity to achieve optimal IFT conditions. Therefore, the presence of alkali plays an important role in ASP flooding.

However, a major challenge with alkali use is the formation of inorganic scale due to reactions with divalent ions like calcium and magnesium in formation water. These reactions can cause deposits of minerals such as calcium carbonate or sulfates, leading to formation damage and operational issues. Scale tends to form near the wellbore or in production facilities. Alkali can also dissolve silica from rock, which becomes unstable when the high-pH ASP fluid mixes with neutral connate water. This pH drop causes the dissolved

silica to polymerize into colloidal particles, as presented in the reactions below, potentially worsening scaling near the well [Arendorf et al., 2011]:



Dimer $\rightarrow$ Cyclic $\rightarrow$ Colloidal $\rightarrow$ Amorphous Silica scale

It is known that the formation of calcium carbonate (CaCO_3) scale is largely determined by pressure, pH, temperature, and the concentrations of calcium and carbonate. Magnesium will precipitate as $\text{Mg}(\text{OH})_2$, and amorphous magnesium silicate will form due to bridging of colloidal silicate particles with magnesium ions. The colloidal silicate particles continue to grow in the absence of divalent ions, resulting in the formation of amorphous silica scale [Arendorf et al., 2011]. The solubility of CaCO_3 decreases with increasing temperature and decreasing pressure, thus resulting in increased scaling tendencies and a potential increase in scale mass.

Risk of Active Aquifer Drift toward ASP Pilot. Historical data indicate that maintaining reservoir pressure of around 3,800 psi in the ASP pilot area requires support from surrounding aquifers. Simulation of tank models shows this pressure cannot be sustained without natural water influx, given limited injection volumes (see Fig. 7). A full-field model reveals strong aquifer connections to the Z46CH sand, particularly from the northeast and southeast, contributing to water inflow as represented in Fig. 8. Meanwhile, fluids primarily exit the pilot zone to the southwest and northwest. This flow pattern means that after soft water is injected during the ASP pre-flush, formation water rich in divalent ions may re-enter the area, potentially causing scaling issues.

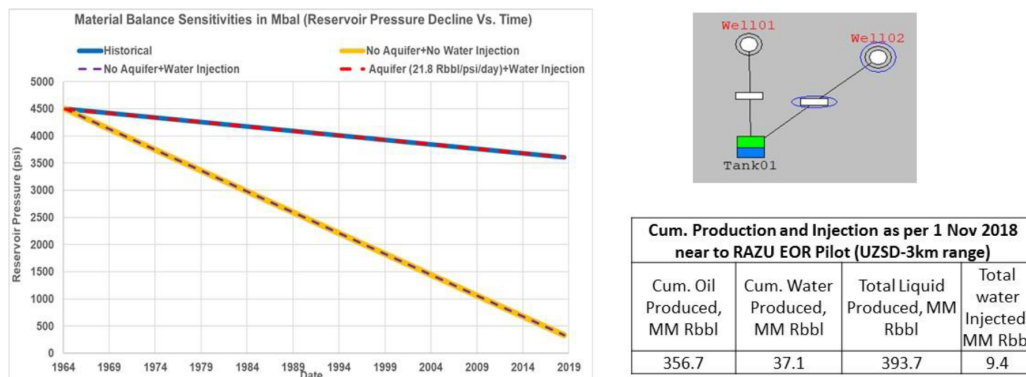


Figure 7—Tank model material balance simulation results using Mbal software

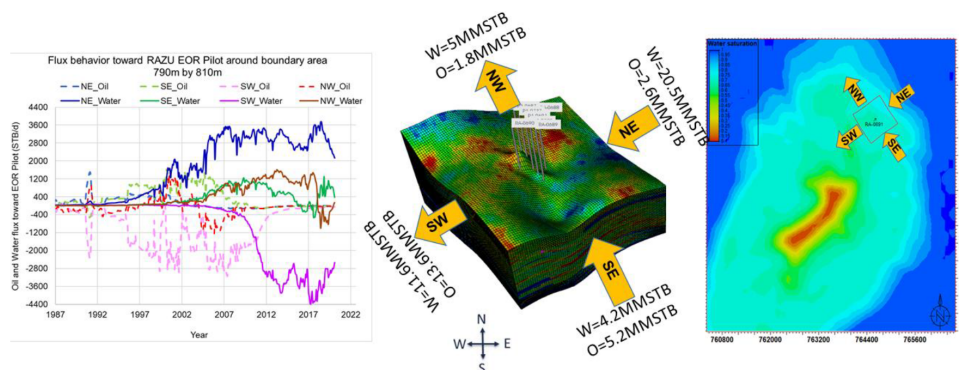


Figure 8—Analysis of influx and outflux in EOR pilot area from history matched full field simulation model

Risk of Ineffective Pre-Flush Performance in Achieving Target Salinity and Hardness. Between September 2023 and July 2025, softened water was injected into the pilot area at variable rates ranging from 500 to 2,000 barrels per day (bbl/day), with corresponding adjustments in injection pressure. The injected fluid consistently met the target specification, with softened water maintained at approximately 30,000 ppm total dissolved solids (TDS) and hardness below 5 ppm. By this period, the cumulative volume of injected softened water had reached approximately three pore volumes (PV), as illustrated in Plot (a) of Fig. 9. The central injection well maintained a steady injection rate of 2,000 bbl/day with relatively stable bottomhole injection pressure after the restart of the injection pilot in December 2025, as depicted in Plot (b) of Fig. 9.

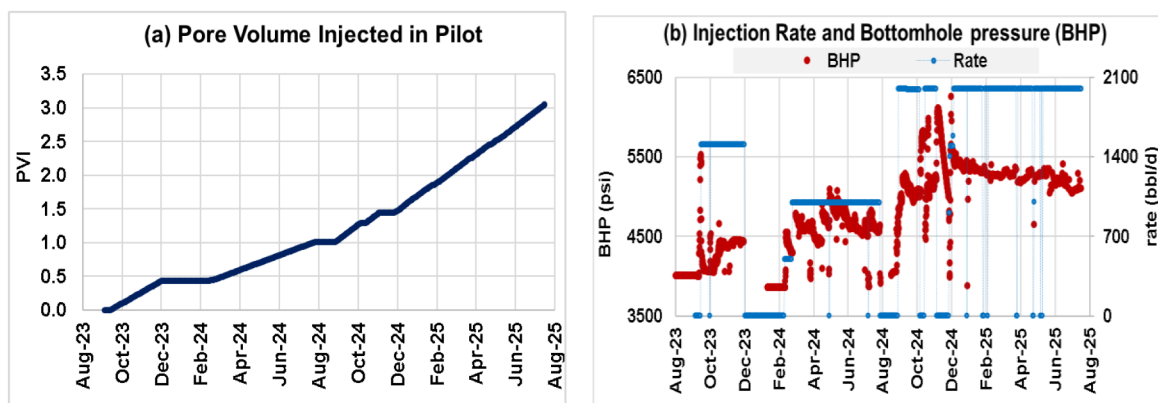


Figure 9—Pilot injection performance analysis

Electrical submersible pump (ESP) sensor data were utilized to assess the reservoir response. An upward trend in pump intake pressure was observed across all producing wells after the restart of softened water injection in December 2025, indicating increasing backpressure from the formation, as shown in Plot (a) of Fig. 10. However, pump intake temperature responses varied: it increased in Producer 1, remained stable in Producer 2, and decreased in the other producers as shown in Plot (b) of Fig. 10, suggesting heterogeneous flow paths and variable injector-to-producer connectivity.

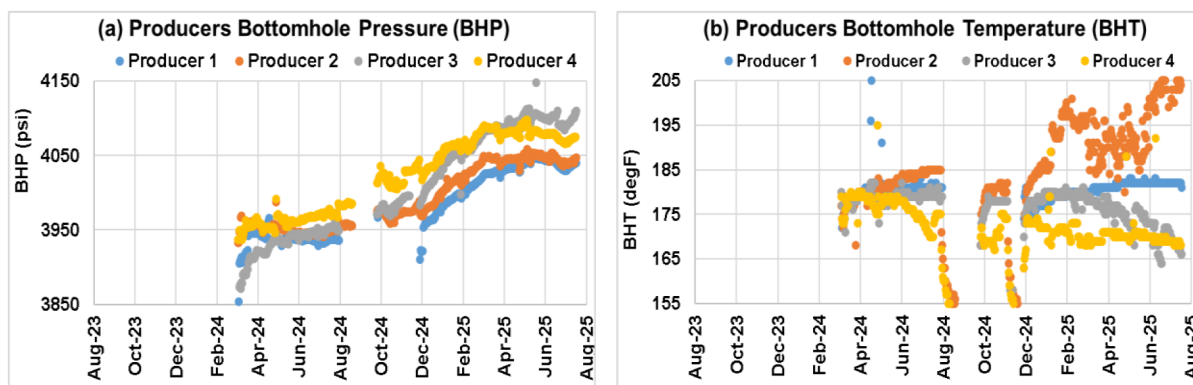


Figure 10—Pilot producers bottom hole pressure and temperature analysis

Following the initiation of water injection, all producers quickly reached nearly 100% water cut {Plot (a) of Fig. 11}, indicating the presence of only residual oil saturation (S_{orw}) within the pilot area. Produced water analysis showed minimal change in salinity and hardness prior to December 2025, but a notable TDS reduction occurred afterward {Plot (b) of Fig. 11}. Producer 4 responded almost immediately with a sharp decline in TDS, while the other producers in the pilot exhibited delayed responses of up to 90 days. Producers 3 and 4, located farther from the aquifer, showed substantial TDS reduction, whereas Producers 1

and 2, positioned nearer to the aquifer, experienced limited changes. Similarly, hardness reductions were significant in Producers 3 and 4 but marginally in the other wells, as presented in Plot (c) of Fig. 11.

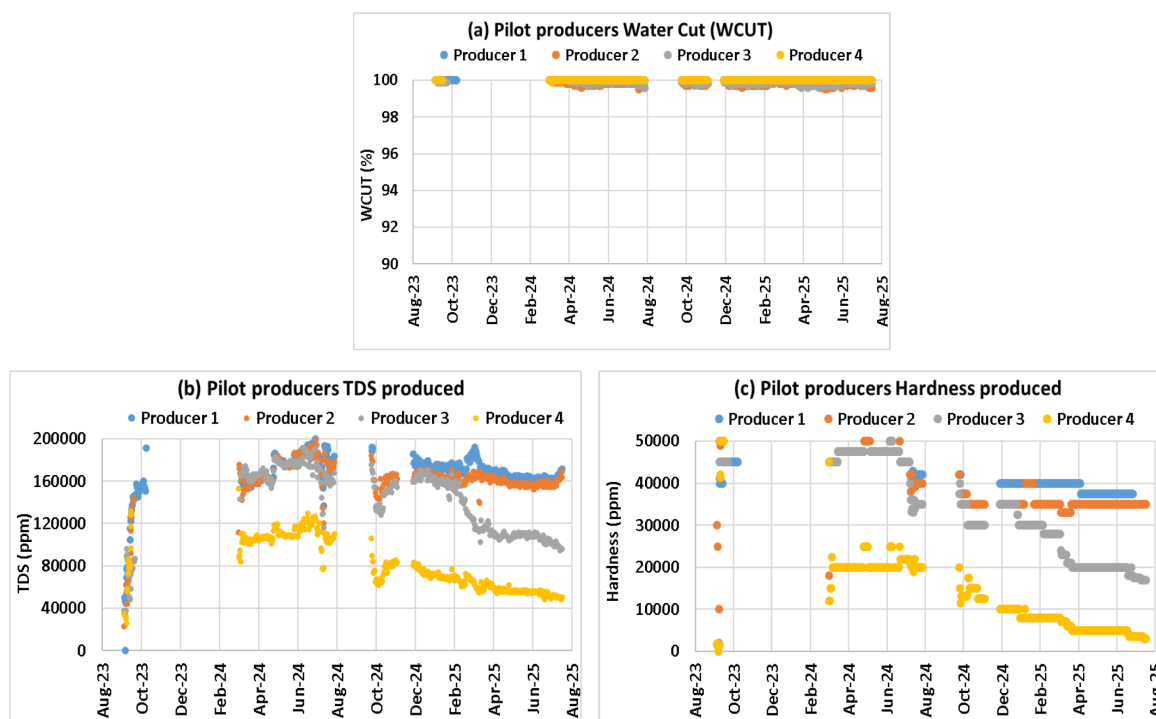


Figure 11—Pilot producers (a) water cut, (b) TDS and (c) hardness production analysis

To better understand flow dynamics, a passive tracer test was conducted by injecting 10 kg of tracer into the central injector on February 10, 2025. Samples were collected from the four pilot producers and three offset wells completed in the same reservoir interval (Z46CH). Tracer breakthroughs were detected at different times, as shown in Fig. 12 i.e., 15 days in Producer 4, 40 days in Producer 3, and 60 days in Producers 1 and 2. No tracer was observed in the offset wells, confirming the confinement of injected fluids within the pilot area. The tracer results, along with salinity and hardness trends, confirmed varying degrees of communication between the injector and producers, with persistent high salinity and divalent ion concentrations observed in wells located close to the aquifer, highlighting uneven displacement efficiency and potential aquifer influence within the EOR pilot zone.

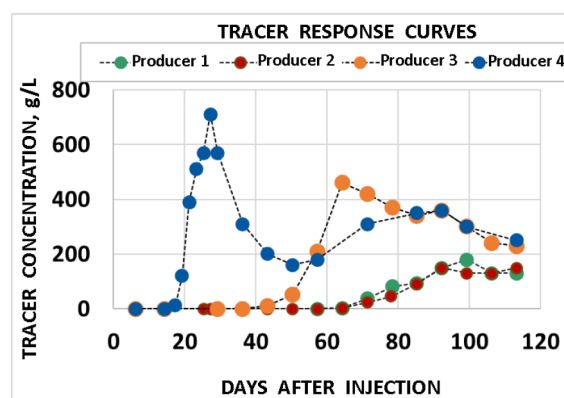


Figure 12—Pilot passive tracer production analysis

ASP Flooding Pilot's performance Comparison with other ASP projects globally. ASP flooding has been implemented in various oil fields worldwide, both at the pilot and commercial development stages. One of the most notable large-scale applications is in the Daqing Oilfield, China, where extensive pilots have been successfully scaled up to full-field operations. These efforts have delivered incremental oil recovery ranging from 20–33% of OOIP, with relatively cost-effective performance at 28–49 \$/bbl incremental oil (Shutang and Qiang, 2010; Guo et al., 2017). In India, the Mangala Oilfield has shown promising results during pilot testing, achieving ~15–25% incremental recovery of OOIP. These encouraging outcomes have prompted further development and scale-up toward field-level ASP implementation (Mandal, 2015). Canada's Taber South Oilfield stands out as an example of full-field ASP deployment. The project demonstrated ~11–16% OOIP incremental recovery, along with a significant boost in production. However, operational challenges—such as scale formation and reduced injectivity—have affected long-term reliability (Ghorbanpour and Khodapanah, 2025). In the United States, ASP pilots have also shown economic viability and technical success, particularly in suitable geological contexts. For example, the Tanner Field in Wyoming achieved an estimated 17% OOIP incremental recovery, with a cost of \$5.85 per barrel, and a payback period of just 2.5 years, based on early 2000s oil prices. Similarly, the Sho-Vel-Tum Field in Oklahoma reported an increase of 10,444 barrels of oil over a 1.3-year pilot period. Other pilot projects conducted in the Minnelusa Formation demonstrated good oil recovery using low surfactant concentrations (~0.1 wt%) (Sheng, 2013).

More recently, Kuwait completed its first ASP pilot in a Sabriyah Maaddud (SAMA) carbonate reservoir worldwide. Despite the reservoir's complex characteristics, presence of high-permeability streaks, the project achieved ~10% OOIP incremental oil recovery. This success was supported by deep conformance control techniques, such as the use of polymer-based organic gels to manage sweep efficiency (Al-Murayri et al., 2022; 2021a; 2021b). However, it is important to note that most of these field trials have been conducted in relatively less harsh reservoir environments. For instance, the SAMA ASP pilot in Kuwait operated under normal injection conditions, without significant aquifer interference. In contrast, the current ASP pilot in Kuwait's RAZU reservoir was designed with an inverted five-spot pattern to reduce chemical losses due to the deeper reservoir depth. Despite this, a strong active aquifer continued to inject high-salinity formation water into the pilot area, limiting the effectiveness of the ASP flood. The inverted pattern (a central injector surrounded by 4 producers) also posed challenges in maintaining sweeping efficiency, as injected soft water could not fully displace the intruding formation water (Khan, 2024). This pilot has challenge of elevated scale precipitation like Canada's Taber South Oilfield.

Risk of polymer viscosity reduction and precipitation. Ward and Martin (1981) present detailed work for the influence of divalent cations on the viscosity of hydrolyzed polyacrylamides polymer. Fig. 13 presents the relationship between fractions of viscosity retained with fractions of divalent ions (Ca^{++} and Mg^{++}) for partially-hydrolyzed-polyacrylamide (HPAM) type polymer. This Figure clearly shows that the increase in hardness can reduce more than twice the viscosity of the polymer. From a practical point of view, the presence of divalent ions also has other drawbacks such as polymer precipitation.

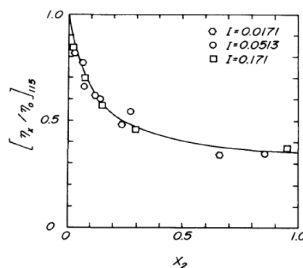


Figure 13—Viscosity reduction due to presence of divalent ions in Hydrolyzed polyacrylamide polymer [Ward and Martin, 1981]

In this Figure, the hardness is represented by the factor X_2 describe as:

$$X_2 = \frac{M_2}{M_1 + M_2}$$

Where M_1 and M_2 are the concentrations (moles/ liter) of the univalent and divalent cations, respectively.

Ineffectiveness of ASP Formulation. An extended pre-flush phase was designed due to active strong aquifer drift toward the EOR pilot. However, injecting large volumes of softened water during this extended pre-flush can cause considerable cooling in the reservoir. Thermal simulations predict temperature drops from 90°C to between 50°C and 70°C within the pilot zone (Fig. 14). This temperature decrease can affect the efficiency of the ASP chemical formulation. Because of this cooling effect, the original ASP formulation tested in the lab as well as in the field through SWCTT was re-evaluated at lower temperatures.

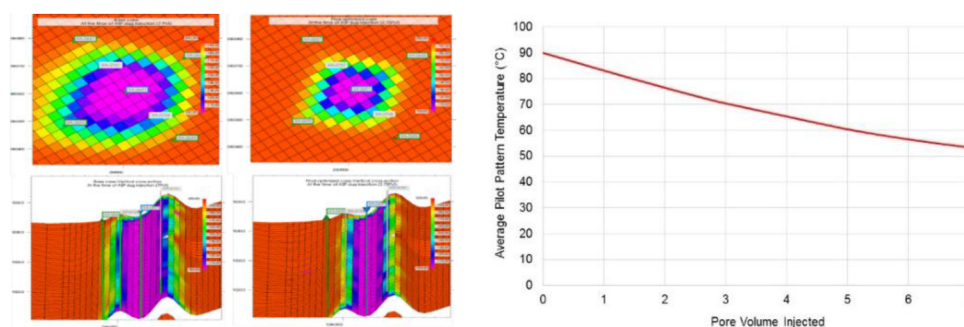


Figure 14—EOR pilot area cooling effect due to pre-flush phase

Core flooding experiments showed a notable decline in oil recovery when the temperature dropped from 90°C (KOC#35) to 50°C (KOC#39), as shown in Fig. 15 and detailed in Table 1. To determine if this decline was linked to changes in optimal salinity, phase behavior tests were performed at 50°C, revealing that optimal salinity shifted to 2 wt.% NaCl. Additional core floods at 2 wt.% NaCl and 90°C (KOC#38) showed similarly reduced oil recovery and production (Fig. 15), confirming that conditions below optimal salinity yield similarly reduced oil recovery.

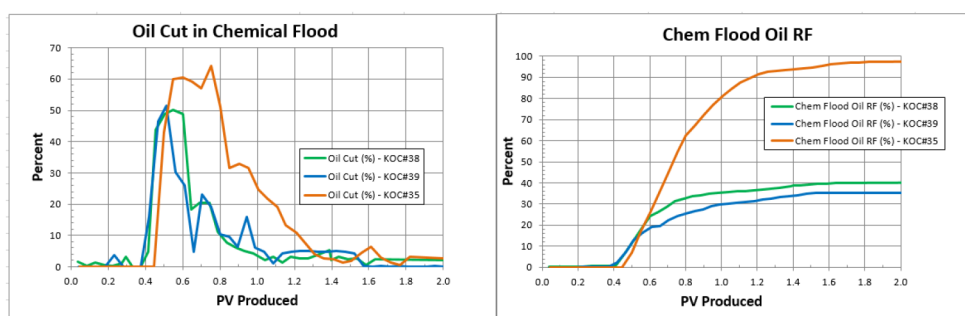


Figure 15—ASP flooding performance in core flood at low temperature (KOC#39: blue line), high temperature (optimal salinity KOC#35: orange line and under-optimal salinity KOC#38: green line) [Al-Murayri et al., 2019a].

Although surfactant concentration in KOC#35 was lower than in KOC#38 and KOC#39, higher surfactant levels would generally boost oil recovery. Co-solvent concentration varied but did not significantly impact results if it prevented surfactant phase separation. Remaining oil saturation was lowest in KOC#35 (0.01), much better than in KOC#39 (0.29) and KOC#38 (0.25), indicating the latter two floods underperformed and further formulation adjustment was necessary for lower temperatures.

Table 1—Core flooding at 90°C and 50°C using original ASP formulation design.

Flood Name	Porous med. & #	Size	PV, cc	Perm., Darcy	Surfactant	NaCl, %	Na ₂ CO ₃ , %	Polymer ATBS (ppm)	Co-solvent, LEA %	Chemical slug size, PV	Soi	Water flood Recovery (%)	Sorw (oil sat. after water flood)	Chemical Flood Recovery (% of Sorw)		Sor* (oil sat. after CF)	Mobilized HC by Chemical (Sorw - Sor*)
														1 PV	2 PV		
KOCH35	BT#27	1.5"x12"	79.4	3.0	0.42% IOS 24-28 0.28% IOS 15-18	3.0	1.5	1500	1.5	0.3	0.77	54.5	0.35	77.1	97.5	0.01	0.34
KOCH38	BT#29	1.5"x12"	83.9	1.9	0.6% IOS 24-28 0.4% IOS 15-18	2.0	1.5	1500	1.2	0.3	0.77	44.8	0.43	35.4	40	0.25	0.17
KOCH39	BT#30	1.5"x12"	80.9	2.9	0.6% IOS 24-28 0.4% IOS 15-18	3.0	1.5	1500	1.0	0.3	0.79	43.7	0.45	29.7	35.4	0.29	0.16

Risks Mitigation

Post-Waterflood Residual Oil Saturation (Sorw) Estimation. The residual oil saturation (Sorw) after waterflooding was evaluated through an integrated interpretation of various datasets, including SCAL results, liquid trapper core measurements, SWCTT, time-lapse saturation logging, downhole fluid analysis, and production data from the pilot wells. During the drilling phase, downhole fluid samples were acquired using a repeat formation tester (RFT) equipped with a fluid analyzer. These samples confirmed the absence of any mobile oil phase within the pilot zone, as illustrated in Plot (a) of Fig. 16. A field-based SWCTT conducted estimated residual oil saturation (S_{orw}) at approximately $24\% \pm 2\%$, which falls on the lower end of the range derived from open-hole log interpretation (20–30%), as shown in Plot (b) of Fig. 16. Time-lapse saturation logs acquired in a fiberglass-cased observation well during September 2023 and July 2024 revealed negligible changes in oil saturation over time and 100% water production from pilot producers, further supporting the interpretation that only residual oil remained {Fig. 17 and plot (a) of Fig. 11}. Subsequently, detailed Dean-Stark analysis of core samples recovered using a liquid trapper coring system yielded Sorw values between 15% and 24%, as presented in Fig. 18. Collectively, the evidence from core, log, tracer, and fluid data consistently indicates that the pilot area contains only post-waterflood residual oil saturation, estimated to lie within the range of 15–24%.

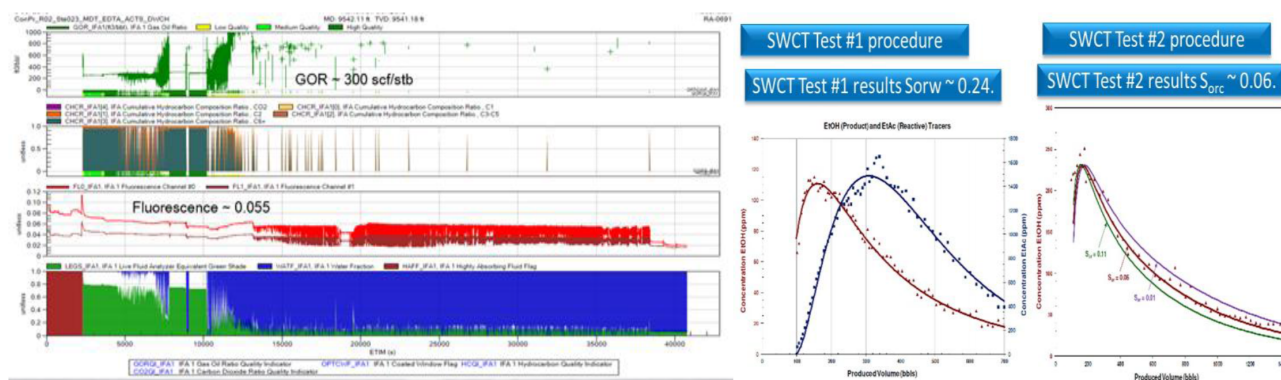


Figure 16—Analyzed downhole sample through downhole fluid analyzer (b) results of SWCTT

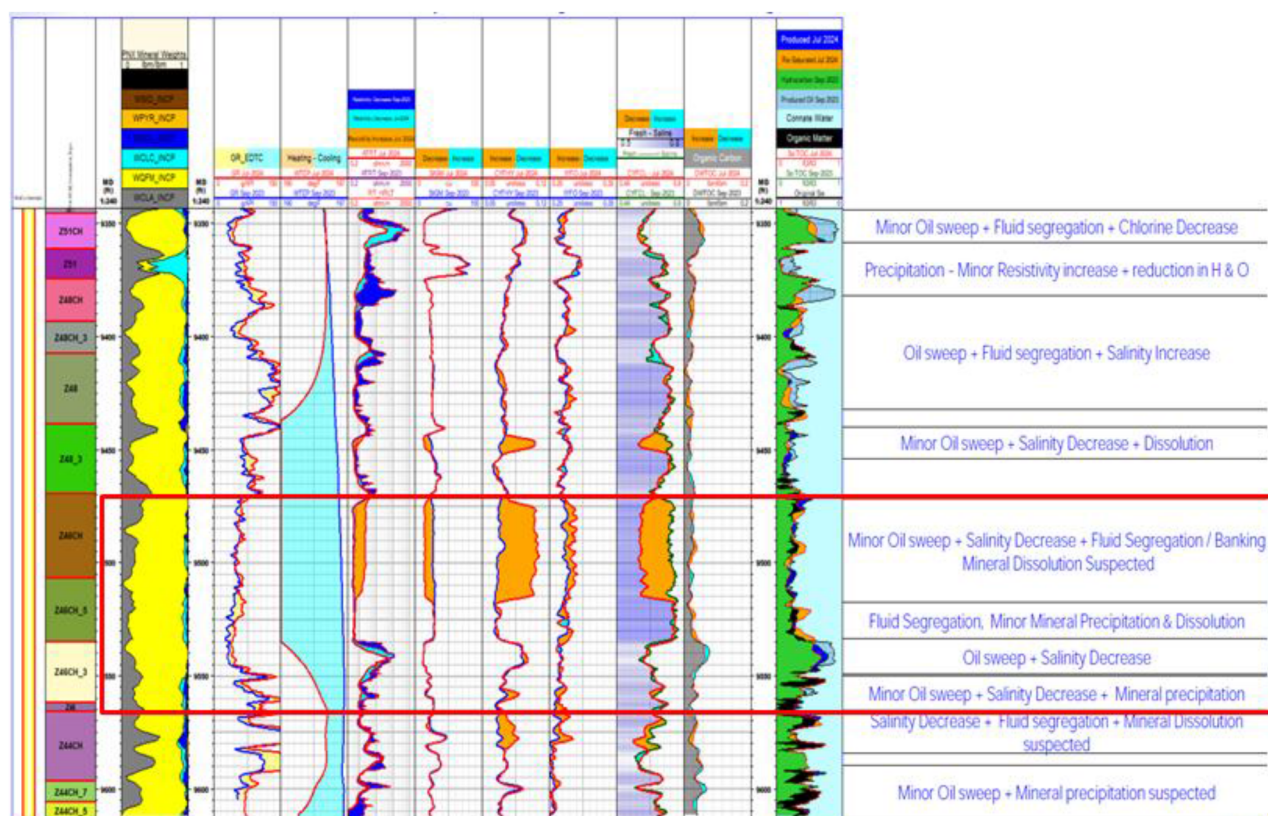


Figure 17—Tim-lapse saturation logs

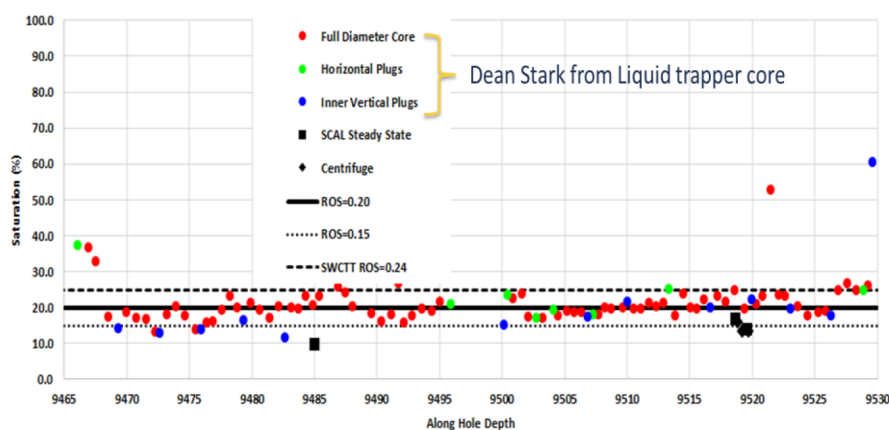


Figure 18—Results of Deak Stark analysis from liquid trapper core taken from one pilot well

Scale Inhibitor, Alkali Type, and Concentration Selection. In the RAZU ASP pilot, both carbonate and silica scale formations are anticipated. Initially, CaCO_3 is expected to dominate, while silica scaling may occur later as ASP concentration in produced fluids increases. Laboratory tests have shown inconsistent results compared to field behavior, highlighting the importance of field-based observations for predicting scaling challenges. Managing scale is a key concern as the project transitions from lab to field. To address this, several strategies are used:

- **Alkali selection** – Weak alkalis like Na_2CO_3 were selected in the ASP formulation design due to their lower tendency to cause scaling compared to strong alkalis like NaOH .
- **Low alkali concentrations:** Operating the pilot at lower pH with low alkali concentrations reduces CaCO_3 risk but can promote silica polymerization, especially below pH 10.5.

- **Use of scale inhibitors:** These prevent or reduce crystal growth and adhesion. Inhibitor selection for RAZU is based on lab tests simulating silica scaling under different fluid mixing scenarios (Wang et al., 2004).

In the RAZU ASP pilot, adding 1.5 wt.% Na_2CO_3 raises surface pH to approximately 11.5, significantly increasing scaling risks due to high calcium concentrations and reservoir temperatures. However, this severe scale precipitation cannot be fully mitigated by currently available scale inhibitors in the market. The estimated window for scale inhibitor application shown here is a simplified guide and does not account for several key factors affecting scale formation. These include alkali breakthroughs in producers, reservoir heterogeneity, deployment methods, artificial lift effects, pressure changes, PVT data, material types, polymer presence, and aquifer influence.

Due to an active aquifer, more formation water is reaching the pilot producers. Moreover, the severe conditions in the RAZU area make it difficult to apply standard inhibitor strategies (Fig. 19). To manage this, it is essential to limit aquifer flow toward the pilot zone. Careful selection of polymers, pre-flush operations, aquifer control measures, and optimized chemical formulations are critical for managing scale risks in such challenging environments.

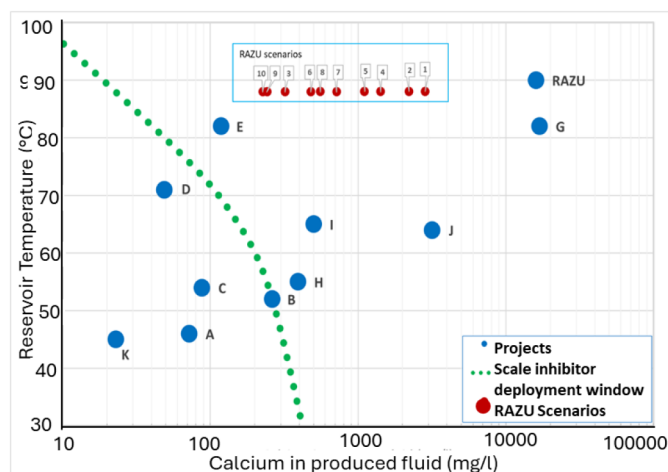


Figure 19—Estimated scale inhibitor deployment window for CaCO_3 scale based on the use of Na_2CO_3 as an alkali in ASP solutions [Al-Murayri et al., 2019b]

Revised and Optimized Pre-Flush Design.

Over-Flush Pre-Flush Strategy

The sector model uses water flux data from the full-field simulation by applying flux boundary conditions predicted from the history-matched full-field model, as shown in Fig. 20. The pilot follows an inverted 5-spot pattern, injecting 1,600 bbl/day centrally and producing 400 bbl/day from each surrounding well during the ASP phase. In a uniform reservoir without aquifer influence, 1–1.5 PVI of pre-flush water would typically prevent scaling and maintain a slight overbalance in VRR, supporting uninterrupted operation. However, due to an active aquifer, a higher pre-flush volume (~2 PVI) is needed to displace formation water from the pilot zone. A strategy involving ~7 PVI of softened water is proposed to mitigate scaling by flushing out divalent-rich water. The influx rates, however, carry significant uncertainty due to the highly permeable reservoir, which can alter pressure gradients over time as production or injection changes in nearby wells. Therefore, the base case was simulated using a chemical EOR model, incorporating aquifer variability ($\pm 50\%$) to assess its impact on ion breakthrough at the four producers (Fig. 21).

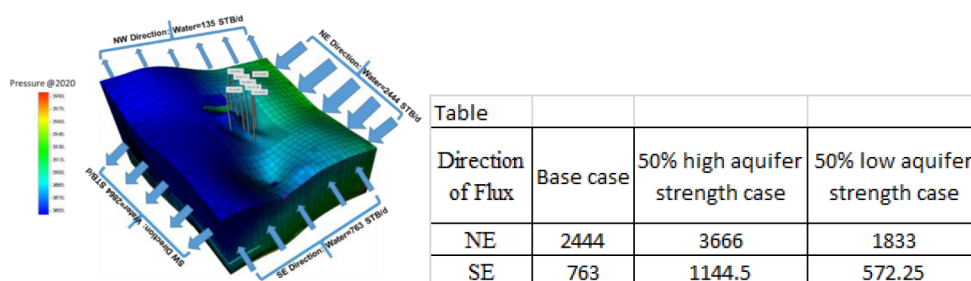


Figure 20—Water influx and out-flux representation in the sector model

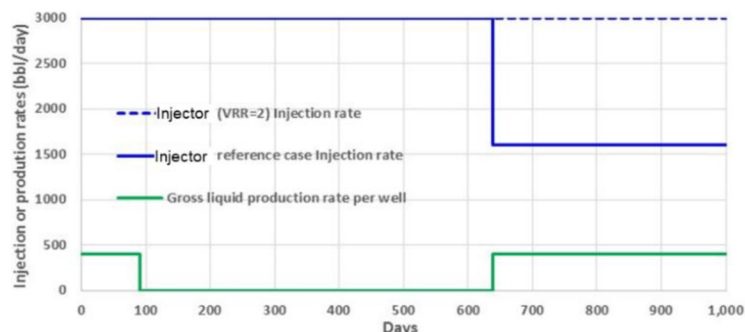


Figure 21—Representation of over flushed pre-flush design

Base case results show that divalent ion production begins to decline around three months after soft water injection starts. However, after injecting 7 PVI and reopening producers, a noticeable reduction in divalent ions was observed only in the no-influx scenario, as shown on Fig. 22. Even in this case, minor ion production ($\sim 1\%$ of formation water) persisted due to local heterogeneities and poor sweep efficiency. In reservoirs with both aquifer support and heterogeneity, a standard pre-flush may not be sufficient to prevent scaling. As a result, further optimization or innovative operational approaches are needed, which will be explored in the following sections.

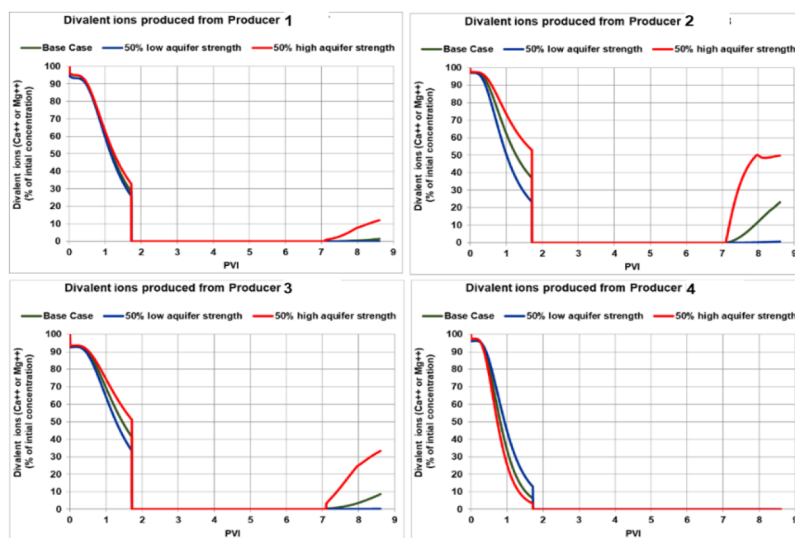


Figure 22—Divalent ions production in base case, 50% high and 50% low aquifer strength cases (Al-Murayri et al., 2019b).

Change of Injection Pattern Strategy

The original pilot design used an inverted 5-spot ASP pattern with a 7 PVI pre-flush, but this approach led to high scaling risk, extended project duration, and increased water treatment costs. To assess alternatives,

a regular 5-spot pattern was tested with two cases: one injecting and producing at 1,600 bbl/day per well, and another at 800 bbl/day. Both followed the same duration as the base case. Results (Fig. 23) showed that higher injection rates reduced divalent ion production more effectively, though about 1% remained. Achieving this reduction required more injection volume, larger facilities, and increased chemical use, raising operational risks and costs. Additionally, most injected chemicals would exit through wells outside the pilot area. If even low levels of divalent ions lead to unmanageable scaling, the pilot may fail to reach its goals. As a result, the regular 5-spot pattern was not recommended for this ASP pilot.

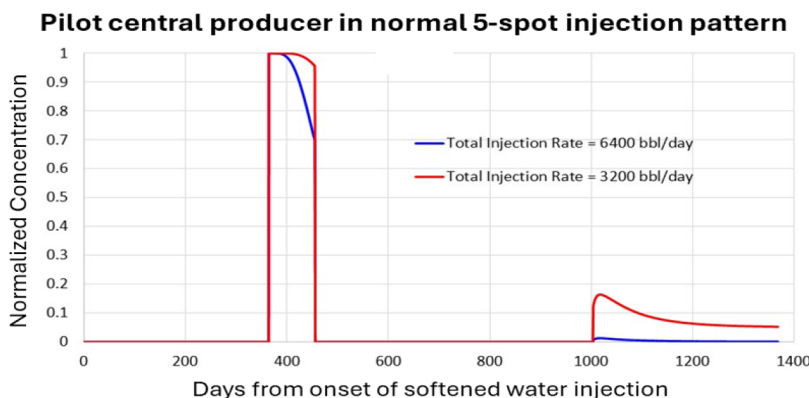


Figure 23—Divalent ions production from by changing injection pattern as regular 5-spot injection pattern (khan et al. 2024)

Optimization of Pre-Flush Strategy

The initial over-flush strategy, involving 7 PVI of softened water, required about 630 days and resulted in significant costs and high divalent ion production (10–22%) in three wells. These ions can react with alkali and form scale, posing operational risks. To address this, pre-flush optimization was performed by incorporating polymer slugs in the pre-flush phase. Multiple scenarios were tested using a chemical EOR simulator, including softened water only (SW), softened water followed by polymer (SW-P), polymer then softened water (P-SW), and combined sequences like SW-P-SW and P-SW-P.

The SW-P scenario was selected as more technically promising because it reduces the risk of polymer degradation by using softened water after polymer injection. Using 3 PVI of softened water followed by 0.8–1 PVI of polymer in the pre-flush phase proved most effective, reducing divalent ion production and pilot duration (Table 2). Other scenarios could also reduce divalent ions similarly but require higher PVI volumes, increasing cost and duration. The optimized case boosts oil recovery and lowers divalent ion levels significantly, down to about 1% of initial concentration, as shown in Fig. 24. The final optimized case reduces pilot duration by 370 days and lowers operating costs by injecting 4.15 PVI of softened water, achieving better oil recovery than the base case (Fig. 24). It also improves chemical efficiency, demonstrated by a higher chemical-to-oil ratio (Fig. 24), and minimizes cooling risk in the pilot area {Plot (d) of Fig. 19}

Table 2—Pre-flush optimization simulation cases descriptions

Sr. No.	Simulation case	Case Type	Simulation case Description	Pre-flush PVI and Time					Max. Divalent ions produced (% of current divalent ions in formation water)
				Polymer slug viscosity (cp)	PVI (P)	PVI (SW)	Total PVI	Time (days)	
1	Base case	SW	7PVI (SW) with Shutin pilot producers	2	0.00	7.00	7.00	630	22
2	Case 1	SW	7PVI (SW) with No Shutin pilot producers	2	0.00	7.00	7.00	630	42
3	Case 2	SW-P	1PVI (SW)+1PVI (P)	2	1.00	1.00	2.00	259	13.6
4	Case 3		2PVI (SW)+1PVI (P)	2	1.00	2.00	3.00	349	8
5	Case 4-Optimised		3PVI (SW)+1PVI (P)	2	1.00	3.00	4.00	439	1
7	Case 5	P-SW	1PVI (P)+1PVI (SW)	2	1.00	1.00	2.00	259	17
8	Case 6		2PVI (P)+1PVI (SW)	2	2.00	1.00	3.00	428	1
9	Case 7		1PVI (SW)+1PVI (P)+1PVI (SW)	2	1.00	2.00	3.00	349	2.4
10	Case 8	SW-P-SW	1PVI (SW)+0.75PVI (P)+1PVI (SW)	2	0.75	2.00	2.75	307	2.2
11	Case 9		1PVI (SW)+0.5PVI (P)+1PVI (SW)	2	0.50	2.00	2.50	265	3.8
12	Case 10		1PVI (SW)+1PVI (P)+2PVI (SW)	2	1.00	4.00	5.00	529	1.9
13	Case 11		1PVI (SW)+0.5PVI (P)+3PVI (SW)	2	0.50	4.00	4.50	445	3
14	Case 12	P-SW-P	1PVI (P)+1PVI (SW)+1PVI (P)	2	2.00	1.00	3.00	428	2
15	Case 13		1PVI (P)+2PVI (SW)+1PVI (P)	2	2.00	2.00	4.00	518	4.2
16	Case 14	SW-P-SW	1PVI (SW)+0.5PVI (P)+1PVI (SW)	8	0.50	2.00	2.50	265	5.00
17	Case 15		0.5PVI (SW)+0.5PVI (P)+0.75PVI (SW)	5	0.50	1.25	1.75	197	9
18	Case 16		1PVI (SW)+0.25PVI (P)+1.5PVI (SW)	5	0.25	2.00	2.25	222	35
19	Case 17		0.5PVI (SW)+0.5PVI (P)+0.75PVI (SW)	10	0.50	1.25	1.75	197	30.00
20	Case 18		1PVI (SW)+0.25PVI (P)+1.5PVI (SW)	10	0.25	2.00	2.25	222	33.00
21	Case 19		0.5PVI (SW)+0.5PVI (P)+0.75PVI (SW)	25	0.50	1.25	1.75	197	30.00
22	Case 20		1PVI (SW)+0.25PVI (P)+1.5PVI (SW)	25	0.25	2.00	2.25	222	
23	Case 21		0.5PVI (SW)+0.5PVI (P)+0.75PVI (SW)	50	0.50	1.25	1.75	197	29.00
24	Case 22		1PVI (SW)+0.25PVI (P)+1.5PVI (SW)	50	0.25	2.00	2.25	222	35.00
25	Case 23		0.5PVI (SW)+0.5PVI (P)+0.75PVI (SW)	75	0.50	1.25	1.75	197	29.00
26	Case 24		1PVI (SW)+0.25PVI (P)+1.5PVI (SW)	75	0.25	2.00	2.25	222	35.00

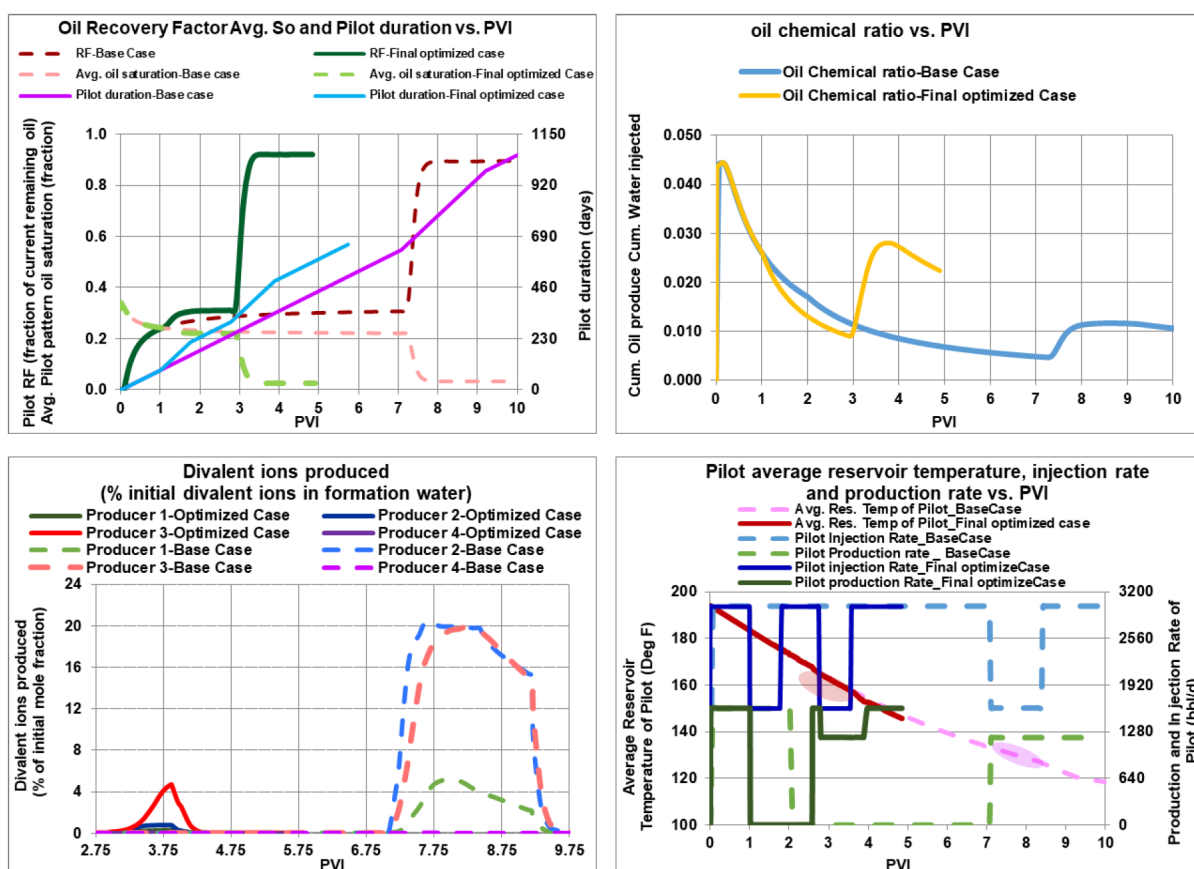


Figure 24—Performance comparison of optimized versus original design pre-flush strategy

Proper Polymer Selection in Harsh Environments. Temperature greatly impacts polymer stability, making chemical flooding more challenging in high-temperature reservoirs. The target reservoir reaches 200°F and contains high levels of divalent ions (15,800 ppm Ca^{2+} , 3,530 ppm Mg^{2+}), increasing the risk of polymer precipitation and viscosity loss due to hydrolysis. These harsh conditions persist even in optimized scenarios. Laboratory results (Fig. 25) show that increasing the ATBS content in polymers raises the cloud

point and improves thermal stability. A polymer with 20% ATBS shows better resistance to degradation. Among tested options, AN125 (an ATBS-based copolymer) offers superior stability compared to FP5115 and FP5220, though at a slightly higher cost. For the RAZU reservoir, a tertiary polymer with ~20% ATBS, ~15–20% AA, and ~60% AM is recommended for enhanced performance and solubility under these extreme conditions.

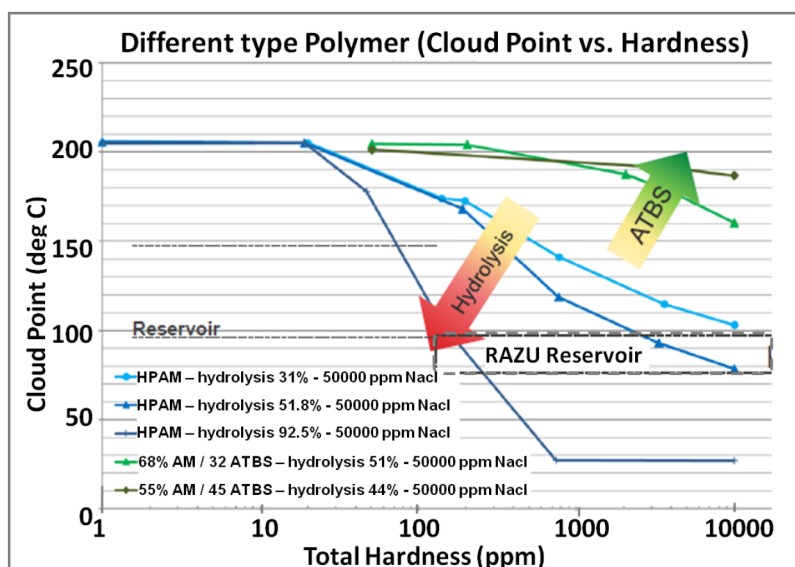


Figure 25—Cloud point temperatures as a function of hardness (in mM) for HPAM and ATBS type polymers [Doe et al., 1987; Moradi-Araghi and Doe, 1987; Moradi-Araghi et al., 1987].

Re-Tuning of ASP Formulation Design. The optimized ASP formulation for the RAZU pilot area incorporates a total of 1 wt% Soloterra surfactant blend to mitigate the cooling effect observed during the pre-flush injection. The final surfactant composition consists of 0.45 wt% Surfactant 1, 0.15 wt% Surfactant 2 (58.3% active content), 0.35 wt% Surfactant 3, and 0.05 wt% Surfactant 4. At 60°C, this formulation exhibits an optimum salinity of 42,500 ppm, achieving a solubilization ratio of 17. The ultra-low IFT behavior, characteristic of a Type-III microemulsion system (Windsor type-III), is observed over a salinity range of 37,000 to 47,000 ppm total TDS. Fig. 26 illustrates the solubilization ratio curve and phase behavior images at 60°C, clearly indicating favorable aqueous solubility.

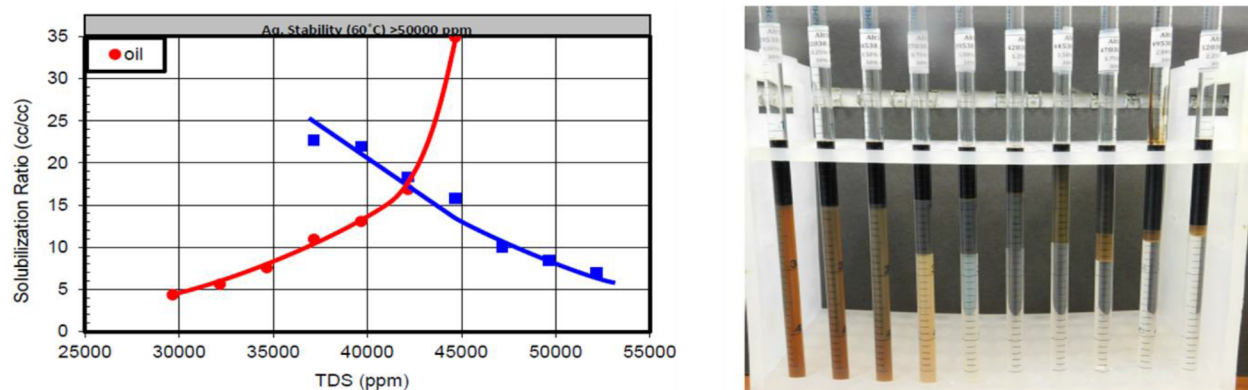


Figure 26—Optimum salinity range to achieve Windsor type-III phase envelope for high solubilization ratio and ultra-low IFT

Following conventional waterflooding, a polymer pre-flush composed of 2,000 ppm sodium carbonate (Na_2CO_3) and 1,000 ppm AN125VHM polymer in SFSB brine was injected, covering 0.25 PV. This was

followed by an ASP slug containing 1.5 wt% Na_2CO_3 , 1.0 wt% surfactant (as described above), and 2,000 ppm AN125VHM polymer, with a slug size of 0.225 PV. A subsequent polymer drive with 2,000 ppm AN125VHM was injected for approximately 3 PV. All fluids were prepared in SFSB brine and injected at a rate of 2.1 ft/day.

Oil saturation was reduced from 34% post-waterflood to just 3% following the ASP injection, indicating residual oil recovery of approximately 92% (see Fig. 27 and Table 3). Out of 156 mg surfactant injected, 67 mg was recovered in the effluent based on measured and interpolated samples (Fig. 28), corresponding to surfactant retention of 0.14 mg/g-rock. When adjusted for interpolation, recovered surfactant increases to 103 mg, reducing retention to 0.09 mg/g-rock. No polymer propagation issues were observed during the pilot.

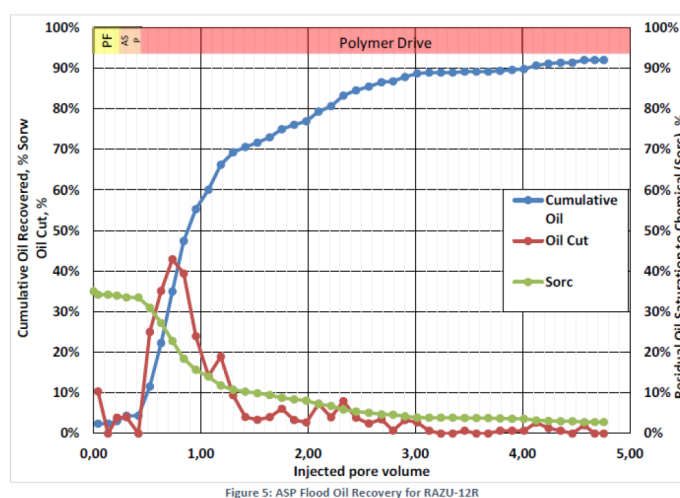


Figure 5: ASP Flood Oil Recovery for RAZU-12R

Figure 27—Oil recovery, oil cut and oil saturation from ASP core flooding

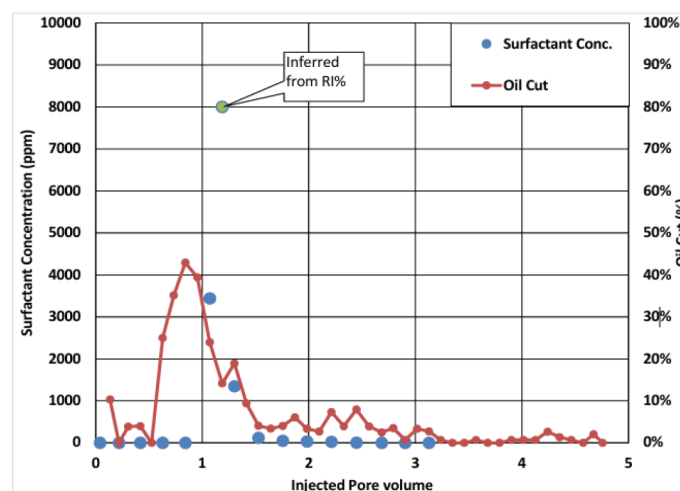


Figure 28—Surfactant production after ASP flooding in core

In conclusion, the optimum ASP recipe consists of 1.5 wt% Na_2CO_3 , 1.0 wt% Soloterra surfactant blend (0.45% Surfactant 1, 0.15% Surfactant 2, 0.35% Surfactant 3, and 0.05% Surfactant 4), and 2,000 ppm AN125VHM polymer. This formulation has demonstrated operational stability and effectiveness across temperatures of 60°C to 90°C.

Table 3—Oil saturation in oil, water and ASP flooding

Oil Flood		Waterflood		ASP Flood	
S_{wr}	0.25	S_w	0.66	S_w	0.97
S_{oi}	0.75	S_{orw}	0.34	S_{orc}	0.03

Lab-scale results were scaled up to field implementation

The ASP formulation was developed based on extensive laboratory studies, as detailed in the section titled Re-Tuning of ASP Formulation Design. The composition includes 1 wt% of a surfactant blend (comprising 0.45 wt% Surfactant 1, 0.15 wt% Surfactant 2, 0.35 wt% Surfactant 3, and 0.05 wt% Surfactant 4), 1.5 wt% sodium carbonate (Na_2CO_3), and 2000 ppm of AN125VHM polymer. This formulation demonstrated strong thermal stability and effectiveness within a temperature range of 60°C to 90°C. It also achieved ultra-low interfacial tension (IFT) at an optimal salinity of 42,500 ppm.

A core flooding experiment was conducted by first injecting 1 Pore Volume Injected (PVI) of softened water at the same salinity, followed by 0.3 PVI of the ASP slug, and finally 1.5 PVI of polymer solution. The results indicated minimal surfactant retention (0.09 mg/g of rock), high oil recovery efficiency (92% of Original Oil in Place, OOIP), and a residual oil saturation of only 3% in the core plug after chemical injection. The same formulation and injection sequence will be applied in the field pilot test, using identical PVI values as those used in the laboratory experiment.

Cost-benefit analysis comparing the original and optimized ASP strategies

The optimized injection strategy (consisting of 2 PVI of softened water, followed by a 0.8 PVI polymer slug and only 1 PVI of chase polymer) offered significant cost saving over the original design (7 PVI of softened water in the pre-flush and 1.8 PVI of chase polymer). This strategy reduced the pilot duration by ~370 days and eliminated 4.3 PVI of softened water injection, resulting in lower injection costs, reduced reservoir cooling, improved ASP formulation efficiency, enhanced oil recovery, and a better chemical-to-oil ratio. Overall, the optimized strategy is projected to save over **USD 2 million**, broken down as follows:

- Approximately **USD 300,000** from eliminating 4.3 PVI of softened water injection.
- Around **USD 1.11 million** from reduced produced water handling costs (based on 2,000 bbl/day over 370 days).
- Additionally, the cost associated with shorter pilot duration and improved efficiency increase the likelihood of pilot success.

Lessons Learned that could benefit future ASP Flooding deployments in Harsh Environment of Reservoir

The ASP flooding pilot conducted in the RAZU reservoir has provided valuable insights that can significantly enhance the performance of future ASP projects, especially under harsh reservoir conditions—such as high-temperature environments with active aquifers and formation water containing high salinity and hardness. The main lessons learned are summarized below:

- **Injection Pattern Selection:** In pilot areas located near active aquifers, implementing a normal 5-spot injection pattern (one central producer surrounded by four injectors) is recommended. This layout helps control aquifer drift of formation water toward the production well and lowers the risk of scale formation.

- **Temperature-Resilient Formulation Design:** The ASP formulation should be designed and tested in the laboratory to maintain effectiveness over a broad temperature range. This is crucial for accounting for potential reservoir cooling resulting from the injection of pre-flush softened water injection.
- **Fracture Pressure Monitoring:** Observations from the current pilot indicate that reservoir cooling has led to a reduction in fracture pressure. Therefore, it is essential to reassess the fracture pressure before chemical injection and ensure that the chemical injection is carried out under matrix conditions to avoid unintentional fracturing.
- **Chemical Stability Under Harsh Conditions:** All chemicals, particularly polymers, should be selected based on their thermal stability and resistance to degradation in high-salinity and high-hardness environments. This ensures that the polymer retains its viscosity and does not undergo hydrolysis or precipitation, which would compromise its performance.
- **Chemical Efficiency:** Weak alkali should be chosen from laboratory, which minimizes scale precipitation risk. Surfactants with low dynamic retention should be selected to maximize sweep efficiency and reduce chemical loss within the reservoir.
- **Optimized Injection Strategy:** The injection sequence and PVI should be optimized to minimize the mobilization of divalent ions, which can contribute to scale formation. Chemical volumes should be calibrated to balance cost-efficiency with effective oil recovery, especially when planning for field-scale expansion.
- **Laboratory Validation Under Realistic Conditions:** Before field deployment, the ASP formulation must undergo core flood testing at both reservoir temperature and lower temperatures expected after soft water pre-flush. This ensures the formulation delivers high recovery under variable field conditions.
- **Aquifer Control in Inverted Patterns:** In cases where an inverted five-spot injection pattern is adopted near an active aquifer, injecting a viscous polymer slug during the pre-flush phase can help restrict unwanted water movement toward production wells.
- **Comprehensive Project Planning:** Successful ASP application requires careful consideration of key design factors, including chemical formulation tailored to reservoir conditions, core flood testing across temperature ranges, and understanding of formation heterogeneity, mineralogy, and fluid chemistry. Early screening of surfactant and alkali compatibility with formation water is essential to minimize scaling and precipitation.
- **Operational Risk Management:** Project planning should also address operational challenges such as managing emulsion formation, preventing chemical degradation under high temperatures, and minimizing surfactant retention, which can significantly impact project performance and economic feasibility.

Conclusions

- Tracer and produced fluid analyses confirmed inconsistent reductions in salinity (31,000 - 1,800 ppm) and hardness (2,800 -34,000 ppm) across producer wells. Elevated salinity (~180,000 ppm) and persistent divalent ion concentrations (~34,000 ppm) were observed near wells affected by aquifer influx.
- Simulation results revealed that the optimized pre-flush strategy consists of ~2 PVI of softened water followed by a 0.8–1 PVI viscous polymer slug prior to ASP injection.
- This optimized strategy significantly reduced scaling risks by lowering $\text{Ca}^{2+}/\text{Mg}^{2+}$ concentrations from 1,580 ppm to a manageable range of 35–158 ppm at the producer wells.
- The strategy also prevented excessive reservoir cooling by maintaining the temperature at ~70°C instead of 55°C and reduced the overall pilot duration by ~370 days compared to the originally

designed pre-flush phase of 7 PVI of softened water. This helped preserve the ASP formulation's effectiveness, enhanced oil recovery, and improved the chemical-to-oil ratio.

- Commercial scale scavengers were carefully selected to prevent unexpected scale precipitation near the wellbore, ensuring flow assurance and protecting well integrity.
- The selected synthetic polymer (AN125), containing approximately 20% ATBS, demonstrated excellent viscosity retention and resistance to hydrolysis, even under harsh conditions (~280000 ppm salinity, 19000 ppm $\text{Ca}^{2+}/\text{Mg}^{2+}$ and 90°C temperature).
- Fine-tuning of salinity, co-solvent levels, and surfactant blend ratios enabled the optimized ASP formulation to perform effectively across a broader reservoir temperature range of 50°C to 90°C.

Acknowledgments

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Nomenclatures / Acronyms

Symbol	Description
θ	Contact angle
μ	Displacing fluid viscosity
ASP	Alkaline surfactant polymer
ATBS	Acrylamide-tertiary-butyl-sulfonic-acid
EOR	Enhanced oil recovery
HPAM	Partially-hydrolyzed-polyacrylamide
IFT	Interfacial tension
M_1 and M_2	Concentrations in moles per liter of the univalent and divalent cations, respectively
N_c	Capillary-number
P	Polymer Injection
ppm	Part per million
PVI	Pore velum injection
SW	Soften water injection
SWCTT	Single well chemical tracer test
TAN	Total acid number
v	Darcy velocity
VRR	Voidage-replacement-ratio
X_2	Hardness
Z46CH	Geological channel sand name
σ	IFT between the displaced and displacing fluid
SGS	Sequential gaussian simulation
IK	Indicator kriging
RFT	Repeat formation tester
Na_2CO_3	Sodium carbonate

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